Learning Check

1. Different isotopes of an element have the same number of _______________ in the nucleus but different numbers of _______________ .
2. Which of the following is *not* true about a nucleus of the isotope ${}^{18}_{8}O$?
   (a) It contains 8 protons.
   (b) It contains 18 nucleons.
   (c) It contains 8 electrons.
   (d) It contains 10 neutrons.
3. (True or False.) Protons can stay close together inside a nucleus because the repulsive electric force between them disappears at short range.

ANSWERS: 1. protons, neutrons **2.** (c) **3.** False

Learning Check

1. Nuclear radiation can be detected by using a _______________ .
2. Of the three types of nuclear decay, only _______________ does not affect the atomic number of the nucleus.
3. (True or False.) Alpha decay causes the atomic number of the nucleus to increase.
4. Radioactive decay is involved in which of the following?
   (a) Keeping Earth's interior hot.
   (b) Making smoke detectors work.
   (c) Supplying energy for interplanetary space probes.
   (d) All of the above.

ANSWERS: 1. Geiger counter **2.** gamma decay **3.** False **4.** (d)

Learning Check

1. A certain radioisotope has a half-life of 3 hours. If there are initially 10,000 nuclei, after 6 hours have elapsed the number of undecayed nuclei will be
   (a) 10,000.
   (b) 5,000.
   (c) 2,500.
   (d) 0.
2. If a piece of wood is suspected to be several thousand years old, its true age can likely be determined by using the isotope _______________ .
3. (True or False.) Earth is so old that all radioisotopes present when it was formed have decayed away.

ANSWERS: 1. (c) **2.** carbon-14 **3.** False

Learning Check

1. (True or False.) It is possible to change a nucleus that is stable into one that is not stable.
2. __________ activation analysis is used to determine which elements are present in a sample.

ANSWERS: 1. True **2.** Neutron

Learning Check

1. The _______________ of a nucleus is a measure of how tightly bound it is.
2. (True or False.) The average mass of each neutron in the nucleus of an iron atom is not the same as the average mass of each neutron in the nucleus of a gold atom.

ANSWERS: 1. binding energy **2.** True

Learning Check

1. Which of the following is *not* the usual outcome of nuclear fission?
   (a) The binding energy per nucleon of the fission fragments is smaller than that of the original nucleus.
   (b) Energy is released.
   (c) Neutrons are released.
   (d) The fission fragments are radioactive.
2. (True or False.) Of the thousands of different isotopes known to exist, only a few can be used to make an atomic bomb.
3. In a nuclear reactor, ______________ are used to limit the number of neutrons passing from one fuel rod to another.
4. (True or False.) In a worst-case accident at a nuclear power plant, the core could explode like an atomic bomb.

ANSWERS: 1. (a) **2.** True **3.** control rods **4.** False

Learning Check

1. What is the origin of solar energy?
2. A ____________ weapon makes use of both nuclear fission and nuclear fusion.
3. Which of the following is *not* a method of producing nuclear fusion?
   (a) Heating a magnetically confined plasma.
   (b) Forcing nuclei together using lasers.
   (c) Cooling nuclei to very cold temperatures.
   (d) Forcing nuclei together with a nuclear fission explosion.
4. (True or False.) Fusion is used to produce the largest nuclei known to exist.

ANSWERS: 1. fusion of hydrogen **2.** thermonuclear **3.** (c) **4.** True